

Module 3: Warehouse Management

1. Introduction

This document outlines the technical and functional specifications for the Warehouse Management module of the Barq/Alinjaz ERP system. This module is critical for managing the lifecycle of materials used in SEC projects, from initial collection from SEC warehouses to final installation, return, or scrapping. The module will provide a centralized system for inventory control, material tracking, and reconciliation against Work Orders, ensuring accuracy and efficiency in material handling.

2. Core Features

The Warehouse Management module will be comprised of several interconnected tabs, each addressing a specific part of the warehouse workflow. The primary features include:

- **Material Receipt & Inventory:** A digitized process for logging materials collected from SEC warehouses, including automated parsing of material receipts.
- **Work Order Association:** Linking all received materials to specific Work Orders for accurate project costing and tracking.
- **Scrap Management:** A systematic process for tracking and managing replaced or old assets designated as scrap.
- **Material Returns:** Facilitating the return of excess or unused materials back to SEC warehouses.
- **Additional Material Requests:** A workflow for requesting and tracking additional materials required for a Work Order beyond the initial plan.
- **Inventory Tracking & Reporting:** Real-time visibility into material stock levels, movements, and comprehensive reporting.

3. Detailed Tab Specifications

3.1. Tab 1: Material Receipts

This tab will serve as the primary interface for the warehouse team to log materials collected from SEC warehouses.

3.1.1. User Interface

A simple, intuitive interface for uploading scanned copies of the "SEC Contractor/User Checklist" (material receipt) documents. The interface will consist of:

- **File Upload Control:** Allowing users to upload one or more PDF or image files of the material receipts.

- **Receipts Grid:** A grid displaying all uploaded and processed material receipts, with the following columns:
 - **Receipt ID:** A unique identifier for each uploaded receipt.
 - **Upload Date:** The date and time of the upload.
 - **Processing Status:** (e.g., "Pending", "Processed", "Error").
 - **Action Buttons:** "View", "Process", "Delete".

3.1.2. Functional Logic

- 1 **Uploading Receipts:** The user uploads the scanned material receipt. The system saves the file and creates a new entry in the receipts grid with a "Pending" status.
- 2 **Processing Receipts:** When the user clicks "Process", the system will use Optical Character Recognition (OCR) to extract the data from the document. The key fields to be extracted are:
 - **Contractor/User ID:** To verify the materials are for Barq/Alinjaz (ID: 1000329).
 - **Date:** The date of material collection.
 - **Material Details (Line Items):** For each item, the system will extract:
 - **Material Number:** (e.g., 908202098)
 - **Material Description:** (e.g., CLAMP,GRD,BRZ,35-70SQMM CU,16MM DIA CWLD)
 - **Unloading Point:** This is a critical field. The system must parse it to extract the Work Order number and type. The format is W<WO_number><WO_type>/<line_item>. For example, from W241027582805/0055, the system should extract:
 - **Work Order Number:** 241027582
 - **Work Order Type:** 805
 - **Quantity:** The quantity of the material collected.
- 3 **Data Validation & Storage:** After extraction, the system will:
 - Validate the extracted Work Order number against the existing Work Orders in the system.
 - For each line item, create a new record in the MaterialInventory table with a status of "In-Stock".
 - Link each material record to the corresponding Work Order.
 - Update the receipt's status to "Processed". If any errors occur (e.g., invalid WO number), the status will be set to "Error" with a descriptive message.

3.2. Tab 2: Inventory Management

This tab provides a comprehensive view of all materials in the warehouse.

3.2.1. User Interface

A filterable and searchable grid displaying all materials in the inventory. The grid will have the following columns:

- **Material Number**
- **Material Description**
- **Work Order Number**
- **Quantity**
- **Status:** (e.g., "In-Stock", "Issued to Site", "Installed", "Returned", "Scrapped")
- **Date Received**
- **Location:** (e.g., "Main Warehouse", "Site X")

3.2.2. Functional Logic

- **Filtering and Searching:** Users can filter the inventory by Material Number, Work Order Number, Status, and Date Range.
- **Status Updates:** The status of materials reflects their physical state in the project lifecycle. The workflow is as follows:

3.1 **In-Stock:** This is the default status for materials logged via the **Material Receipts** tab. It indicates the materials are physically in the contractor's warehouse.

3.2 **Issued to Site:** The warehouse team will manually change the status to "Issued to Site" when materials are dispatched to the project location. This action creates a record for internal tracking.

3.3 **Installed:** The status is updated to "Installed" based on progress reported in the **Work Order Management** module (specifically, the Electrical and Civil tabs), confirming the material has been used.

3.4 **Returned / Scrapped:** The status is updated via the **Material Returns** or **Scrap Management** tabs respectively.

3.3. Tab 3: Scrap Management

This tab is for managing materials that are old, replaced, or damaged and need to be registered as scrap.

3.3.1. User Interface

A form to register scrap materials, with the following fields:

- **Work Order Number:** (Dropdown of active Work Orders)
- **Material Number:** (Dropdown of materials associated with the selected Work Order)
- **Quantity:** The quantity to be scrapped.
- **Reason for Scrapping:** (e.g., "Replaced", "Damaged", "Old Asset")
- **Photos:** Ability to upload photos of the scrap material.

3.3.2. Functional Logic

- 4 **Registering Scrap:** The user fills out the form and submits it.
- 5 **Inventory Update:** The system updates the status of the corresponding material in the MaterialInventory table to "Scrapped".
- 6 **Scrap Record:** A new record is created in the ScrapLog table.

3.4. Tab 4: Material Returns

This tab manages the process of returning excess materials to the SEC warehouse.

3.4.1. User Interface

A form for creating a Material Return Note, with the following fields:

- **Work Order Number:** (Dropdown of completed or nearly completed Work Orders)
- **Materials to Return:** A grid of materials associated with the selected Work Order, with a field to enter the quantity to be returned.

3.4.2. Functional Logic

- 7 **Creating a Return Note:** The user selects the Work Order and specifies the quantities of materials to be returned.
- 8 **Inventory Update:** The system updates the status of the returned materials to "Returned" in the MaterialInventory table.
- 9 **Return Note Generation:** The system generates a printable Material Return Note to be taken to the SEC warehouse.

3.5. Tab 5: Additional Material Requests

This tab handles requests for additional materials needed for a Work Order.

3.5.1. User Interface

A form for requesting additional materials:

- **Work Order Number:** (Dropdown of active Work Orders)
- **Materials to Request:** A grid to add materials and specify quantities.
- **Justification:** A text field to explain why the additional materials are needed.

3.5.2. Functional Logic

- 10 **Submitting a Request:** The user submits the request.
- 11 **Approval Workflow:** The request goes through an approval workflow (e.g., Project Manager approval).

- 12 **SEC Request:** Once approved, the request is formally submitted to SEC. The system should track the status of this external request.
- 13 **Receiving Materials:** When the additional materials are received, they are logged through the **Material Receipts** tab, and the system will link them to the original request.

4. Database Schema

Below are the proposed new tables for the Warehouse Management module. Existing tables like WorkOrders will be updated with foreign key relationships.

MaterialReceipts

Column Name	Data Type	Description
<u>ReceiptID</u>	INT (PK)	Primary Key
<u>UploadDate</u>	DATETIME	Date of upload
<u>FilePath</u>	VARCHAR(255)	Path to the scanned receipt file
<u>Status</u>	VARCHAR(50)	Processing status

MaterialInventory

Column Name	Data Type	Description
<u>MaterialInventoryID</u>	INT (PK)	Primary Key
<u>WorkOrderID</u>	INT (FK)	Foreign Key to <u>WorkOrders</u> table
<u>MaterialNumber</u>	VARCHAR(100)	SEC Material Number
<u>MaterialDescription</u>	TEXT	Material Description
<u>Quantity</u>	INT	Quantity of the material

Column Name	Data Type	Description
<u>Status</u>	VARCHAR(50)	In-Stock, Issued to Site, Installed, Returned, Scrapped
<u>DateReceived</u>	DATE	Date the material was received
<u>ReceiptID</u>	INT (FK)	Foreign Key to <u>MaterialReceipts</u> table

ScrapLog

Column Name	Data Type	Description
<u>ScrapLogID</u>	INT (PK)	Primary Key
<u>MaterialInventoryID</u>	INT (FK)	Foreign Key to <u>MaterialInventory</u> table
<u>Quantity</u>	INT	Quantity scrapped
<u>Reason</u>	TEXT	Reason for scrapping
<u>DateScrapped</u>	DATE	Date of scrapping

MaterialReturns

Column Name	Data Type	Description
<u>ReturnID</u>	INT (PK)	Primary Key
<u>WorkOrderID</u>	INT (FK)	Foreign Key to <u>WorkOrders</u> table
<u>ReturnDate</u>	DATE	Date of return

MaterialReturnItems

Column Name	Data Type	Description
<u>ReturnItemID</u>	INT (PK)	Primary Key
<u>ReturnID</u>	INT (FK)	Foreign Key to <u>MaterialReturns</u> table
<u>MaterialInventoryID</u>	INT (FK)	Foreign Key to <u>MaterialInventory</u> table
<u>Quantity</u>	INT	Quantity returned

AdditionalMaterialRequests

Column Name	Data Type	Description
<u>RequestID</u>	INT (PK)	Primary Key
<u>WorkOrderID</u>	INT (FK)	Foreign Key to <u>WorkOrders</u> table
<u>RequestDate</u>	DATE	Date of request
<u>Justification</u>	TEXT	Justification for the request
<u>Status</u>	VARCHAR(50)	Pending, Approved, Rejected, Fulfilled

AdditionalMaterialRequestItems

Column Name	Data Type	Description
<u>RequestItemID</u>	INT (PK)	Primary Key
<u>RequestID</u>	INT (FK)	Foreign Key to <u>AdditionalMaterialRequests</u> table
<u>MaterialNumber</u>	VARCHAR(100)	SEC Material Number
<u>Quantity</u>	INT	Quantity requested

5. Integration Points

- **Work Order Management Module:** The Warehouse module will be tightly integrated with the Work Order module. Specifically, Tab 4 (Material Collection) of the Work Order module will now be driven by the data from the Warehouse module. When materials are assigned to a Work Order in the warehouse, they will appear in the Material Collection tab for the field teams.
- **HR Module:** The system can be extended to track which warehouse staff handled which materials, integrating with the employee data from the HR module.

6. Non-Functional Requirements

- **Performance:** The OCR processing of material receipts should be asynchronous to avoid blocking the UI. The user should be notified when the processing is complete.
- **Usability:** The interface should be designed for quick and easy data entry, especially for the warehouse staff who will be using it daily.
- **Security:** Access to the Warehouse Management module should be role-based. Only authorized warehouse personnel should be able to add or modify inventory data.